

TD-4 Analysis

Exercise 1

(a) Riemann Sums for $f(x) = x^2$ over $[0, 2]$ with $n = 4$

Partition the interval

$$\Delta x = \frac{2 - 0}{4} = 0.5$$

Partition points: $x_0 = 0, x_1 = 0.5, x_2 = 1, x_3 = 1.5, x_4 = 2$.

Compute Riemann sums

$$L_n = \Delta x [f(x_0) + f(x_1) + \cdots + f(x_{n-1})]$$

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Function: f(x) = x^2, Interval: [0.0, 2.0], n = 4
Riemann Partitions [L]: |-----|-----|-----|-----|
Riemann Partitions [R]:  -----|-----|-----|-----|
Riemann Partitions [M]: *--|--*--|--*--|--*--|--*
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- Left Riemann Sum (LRS)

$$\begin{aligned} LRS &= 0.5 [f(0) + f(0.5) + f(1) + f(1.5)] \\ &= 0.5 [0 + 0.25 + 1 + 2.25] = 1.75 \end{aligned}$$

- Right Riemann Sum (RRS):

$$\begin{aligned} RRS &= 0.5 [f(0.5) + f(1) + f(1.5) + f(2)] \\ &= 0.5 [0.25 + 1 + 2.25 + 4] = 3.75 \end{aligned}$$

- Midpoint Riemann Sum (MRS)

$$MRS = 0.5 [f(0.25) + f(0.75) + f(1.25) + f(1.75)]$$

$$= 0.5 [0.0625 + 0.5625 + 1.5625 + 3.0625] = 2.625$$

Exact integral

$$\int_0^2 x^2 dx = \left. \frac{x^3}{3} \right|_0^2 = \frac{8}{3} \approx 2.6667$$

Comparison:

- LRS= 1.75 (underestimates)
- RRS= 3.75 (overestimates)
- MRS= 2.625 (close to exact $2.\bar{6}$)

(b) Right Riemann Sum for $\int_1^4 (3x + 1) dx$ with $n = 6$

Partition the interval

$$\Delta x = \frac{4 - 1}{6} = 0.5$$

Partition points: $x_0 = 1, x_1 = 1.5, \dots, x_6 = 4$.

Right Riemann Sum (RRS)

$$\begin{aligned} RRS &= 0.5 [f(1.5) + f(2) + f(2.5) + f(3) + f(3.5) + f(4)] \\ &= 0.5 [5.5 + 7 + 8.5 + 10 + 11.5 + 13] = 27.75 \end{aligned}$$

Exact integral

$$\begin{aligned} \int_1^4 (3x + 1) dx &= \left(\frac{3x^2}{2} + x \right) \Big|_1^4 \\ &= (24 + 4) - (1.5 + 1) = 25.5 \end{aligned}$$

RRS= 27.75 overestimates the exact 25.5.

(c) Midpoint Riemann Sum for $\int_0^1 e^x dx$ with $n = 5$

Partition the interval

$$\Delta x = \frac{1-0}{5} = 0.2$$

Midpoints: $x_k^* = 0.2k - 0.1$ for $k = 1, \dots, 5$

Step 2: Exact Midpoint Riemann Sum

$$MRS = 0.2 \sum_{k=1}^5 e^{0.2k-0.1} = 0.2e^{-0.1} \sum_{k=1}^5 (e^{0.2})^k$$

Simplify using geometric series

$$= 0.2e^{-0.1} \left(\frac{e^{0.2}(e^1 - 1)}{e^{0.2} - 1} \right) = \frac{e^{1.1} - e^{0.1}}{5(e^{0.2} - 1)}$$

Exact value

$$\int_0^1 e^x dx = e - 1$$

(e) Exact Value of $\int_0^3 (2x + 1)dx$ via Riemann Sum

Partition and points

$$\Delta x = \frac{3}{n}$$

$$x_k = \frac{3k}{n} \text{ for } k = 1, \dots, n$$

Riemann sum formula

$$R_n = \sum_{k=1}^n f(x_k) \Delta x = \frac{3}{n} \sum_{k=1}^n \left(2 \cdot \frac{3k}{n} + 1 \right)$$

Simplify the sum

$$\begin{aligned} &= \frac{3}{n} \left(\frac{6}{n} \sum_{k=1}^n k + \sum_{k=1}^n 1 \right) \\ &= \frac{3}{n} \left(\frac{6}{n} \cdot \frac{n(n+1)}{2} + n \right) \\ &= \frac{3}{n} (3n + 3 + n) = 12 + \frac{9}{n} \end{aligned}$$

Take limit

$$\int_0^3 (2x + 1)dx = \lim_{n \rightarrow \infty} \left(12 + \frac{9}{n} \right) = 12$$

Verification

$$\int (2x + 1)dx = x^2 + x + C$$

Evaluated from 0 to 3

$$(9 + 3) - (0 + 0) = \boxed{12}$$

(f) Left Riemann Sum Approximation of $\int_1^2 \frac{1}{x} dx$ with $n = 4$

Partition the interval

$$\Delta x = \frac{2-1}{4} = \frac{1}{4}$$

Partition points: $x_k = 1 + \frac{k}{4}$ for $k = 0, 1, 2, 3, 4$

Left Riemann Sum (LRS)

$$\begin{aligned} LRS &= \Delta x \sum_{k=0}^3 f(x_k) = \frac{1}{4} \sum_{k=0}^3 \frac{1}{1 + \frac{k}{4}} \\ &= \frac{1}{4} \left(\frac{1}{1} + \frac{1}{1.25} + \frac{1}{1.5} + \frac{1}{1.75} \right) \\ &= \frac{1}{4} \left(1 + \frac{4}{5} + \frac{2}{3} + \frac{4}{7} \right) \end{aligned}$$

Exact Value

$$\int_1^2 \frac{1}{x} dx = \ln(2)$$

Final Form:

LRS

$$= \frac{1}{4} \left(1 + \frac{4}{5} + \frac{2}{3} + \frac{4}{7} \right)$$

Exact = $\ln(2)$

Note

The left Riemann sum can be written more compactly as

$$LRS = \frac{1}{4} \sum_{k=0}^3 \frac{4}{4+k} = \sum_{k=0}^3 \frac{1}{4+k}$$

Exercise 2

(a)

$$u_n = n \sum_{k=0}^{n-1} \frac{1}{k^2 + n^2}$$

Rewrite as Riemann Sum

$$u_n = \frac{1}{n} \sum_{k=0}^{n-1} \frac{1}{\left(\frac{k}{n}\right)^2 + 1}$$

Take limit as integral

$$\lim_{n \rightarrow \infty} u_n = \int_0^1 \frac{1}{x^2 + 1} dx$$

Evaluate integral

$$\int \frac{1}{x^2 + 1} dx = \arctan(x)$$

$$\int_0^1 \frac{1}{x^2 + 1} dx = \arctan(1) - \arctan(0) = \frac{\pi}{4} - 0 = \boxed{\frac{\pi}{4}}$$

(b) $\ln v_n = \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \left(\frac{k}{n} \right)^2 \right)$

1. Recognize the Riemann sum:

$$\lim_{n \rightarrow \infty} \ln v_n = \int_0^1 \ln(1 + x^2) dx$$

2. Compute the integral using integration by parts:

$$\begin{aligned} \int \ln(1 + x^2) dx &= x \ln(1 + x^2) - \int \frac{2x^2}{1 + x^2} dx \\ &= x \ln(1 + x^2) - 2 \int \left(1 - \frac{1}{1 + x^2} \right) dx \\ &= x \ln(1 + x^2) - 2x + 2 \arctan x + C \end{aligned}$$

3. Evaluate from 0 to 1:

$$x \ln(1 + x^2) - 2x + 2 \arctan x \Big|_0^1 = \ln 2 - 2 + \frac{\pi}{2}$$

4. Exponentiate to find the limit:

$$\lim_{n \rightarrow \infty} v_n = e^{\ln 2 - 2 + \pi/2} = 2e^{\pi/2 - 2}$$

Final Answer

$$\boxed{2e^{\frac{\pi}{2} - 2}}$$

(c)

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{4n^2 + k^2}$$

1. Rewrite the sum

$$\frac{n}{4n^2 + k^2} = \frac{1}{n} \cdot \frac{1}{4 + \left(\frac{k}{n}\right)^2}$$

2. Express as Riemann sum

$$\sum_{k=1}^n \frac{1}{n} \cdot \frac{1}{4 + \left(\frac{k}{n}\right)^2}$$

3. Identify Riemann sum components

- Partition interval $[0, 1]$ into n equal subintervals
- $\Delta x = \frac{1}{n}$ (width of each subinterval)
- Sample points $x_k = k/n$ (right endpoints)
- Integrand function $f(x) = 1/(4 + x^2)$

4. Take the limit

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k) \Delta x = \int_0^1 \frac{1}{4+x^2} dx$$

5. Compute the integral step-by-step

a. Recognize standard form:

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

b. For our case ($a=2$):

$$\int \frac{1}{4 + x^2} dx = \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$$

6. Evaluate definite integral

$$\left. \frac{1}{2} \arctan\left(\frac{x}{2}\right) \right|_0^1 = \frac{1}{2} \left(\arctan\left(\frac{1}{2}\right) - \arctan(0) \right)$$

7. Simplify

$$\frac{1}{2} \arctan\left(\frac{1}{2}\right) - 0 = \frac{1}{2} \arctan\left(\frac{1}{2}\right)$$

Final Answer

$$\boxed{\frac{1}{2} \arctan\left(\frac{1}{2}\right)}$$

Exercise 3

(a)

$$\int_{-1}^1 \frac{x^3}{\sqrt{1-x^2}} dx$$

1) Let $f(x) = \frac{x^3}{\sqrt{1-x^2}}$

2) Check if odd:

$$f(-x) = \frac{(-x)^3}{\sqrt{1-(-x)^2}} = \frac{-x^3}{\sqrt{1-x^2}} = -f(x)$$

3) Since $f(x)$ is odd and limits are symmetric ($-a$ to a) why ?

$$\int_{-t}^t f(x) dx = \int_{-t}^0 f(x) dx + \int_0^t f(x) dx$$

$$\int_{-t}^0 f(x) dx = - \int_t^0 f(-x) dx = \int_0^t f(-x) dx$$

$$\int_0^t f(-x) dx = - \int_0^t f(x) dx$$

$$\int_{-t}^t f(x) dx = - \int_0^t f(x) dx + \int_0^t f(x) dx = 0$$

$$\int_{-1}^1 f(x) dx = 0$$

Final answer

0
(b)

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$$

Let

$$I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$$

Substitute $x = \pi - u$:

$$I = \int_\pi^0 \frac{(\pi - u) \sin u}{1 + \cos^2 u} (-du) = \int_0^\pi \frac{(\pi - u) \sin u}{1 + \cos^2 u} du$$

Observe:

$$I = \pi \int_0^\pi \frac{\sin u}{1 + \cos^2 u} du - \int_0^\pi \frac{u \sin u}{1 + \cos^2 u} du$$

$$\S I = \pi \int_0^\pi \frac{\sin u}{1 + \cos^2 u} du - I$$

Solve for I :

$$2I = \pi \int_0^\pi \frac{\sin u}{1 + \cos^2 u} du$$

$$I = \frac{\pi}{2} \int_0^\pi \frac{\sin u}{1 + \cos^2 u} du$$

Evaluate integral:

Let $v = \cos u$, $dv = -\sin u du$

$$\int_0^\pi \frac{\sin u}{1 + \cos^2 u} du = - \int_1^{-1} \frac{dv}{1 + v^2} = \int_{-1}^1 \frac{dv}{1 + v^2} = \arctan v \Big|_{-1}^1 = \frac{\pi}{2}$$

Finalize:

$$I = \frac{\pi}{2} \cdot \frac{\pi}{2} = \frac{\pi^2}{4}$$

Final Answer

$\frac{\pi^2}{4}$

second method (Ghoumari | Djawad)

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$$

Let $u = x$, $dv = \frac{\sin x}{1 + \cos^2 x} dx$

Then $du = dx$, and for v :

$$v = \int \frac{\sin x}{1 + \cos^2 x} dx = -\arctan(\cos x) + C$$

Apply integration by parts:

$$\int_0^\pi u dv = uv \Big|_0^\pi - \int_0^\pi v du$$

Evaluate boundary term:

$$-x \arctan(\cos x) \Big|_0^\pi = \frac{\pi^2}{4}$$

The remaining integral:

$$\int_0^{\pi} \arctan(\cos x) dx = 0 \text{ (by symmetry)}$$

Combine results:

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4} + 0 = \frac{\pi^2}{4}$$

Final Answer

$$\boxed{\frac{\pi^2}{4}}$$

(c)

$$\int_0^{\pi/2} \ln(\sin x) dx$$

Let

$$I = \int_0^{\pi/2} \ln(\sin x) dx$$

Substitute $x = \pi/2 - t$:

$$I = \int_0^{\pi/2} \ln(\cos x) dx$$

Add original and transformed:

$$2I = \int_0^{\pi/2} \ln(\sin x \cos x) dx$$

Use $\sin x \cos x = \frac{1}{2} \sin 2x$:

$$2I = -\frac{\pi}{2} \ln 2 + \frac{1}{2} \int_0^{\pi} \ln(\sin u) du$$

By symmetry:

$$\int_0^{\pi} \ln(\sin u) du = 2I$$

Solve:

$$2I = -\frac{\pi}{2} \ln 2 + I$$

$$I = -\frac{\pi}{2} \ln 2$$

Final Answer:

$$\boxed{-\frac{\pi}{2} \ln 2}$$

(d)-Showing that:

$$\int_0^1 \frac{\ln(1+x)}{x} dx = \frac{\pi^2}{12}.$$

1. Expand the Integrand Using a Series

We start by recalling the Taylor series expansion for $\ln(1+x)$ around $x = 0$:

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}, \quad \text{for } |x| < 1.$$

Since the interval of integration is $x \in [0, 1]$, this expansion is valid.

Divide both sides by x :

$$\frac{\ln(1+x)}{x} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{n-1}}{n}.$$

2. Integrate Term by Term

Now, integrate the series term by term from 0 to 1:

$$\int_0^1 \frac{\ln(1+x)}{x} dx = \sum_{n=1}^{\infty} (-1)^{n+1} \int_0^1 \frac{x^{n-1}}{n} dx.$$

The integral of x^{n-1} is straightforward:

$$\int_0^1 x^{n-1} dx = \frac{x^n}{n} \Big|_0^1 = \frac{1}{n}.$$

Thus, the expression becomes:

$$\int_0^1 \frac{\ln(1+x)}{x} dx = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2}.$$

3. Relate the Series to the Basel Problem

The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is known as the **Basel problem**, and its value is:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The alternating version of this series is:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.$$

This can be derived by considering the difference between the Basel series and its even terms:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} - \sum_{k=1}^{\infty} \frac{1}{(2k)^2} = \frac{\pi^2}{8} - \frac{\pi^2}{24} = \frac{\pi^2}{12}.$$

4. Final Result

Substituting the series back into the integral:

$$\int_0^1 \frac{\ln(1+x)}{x} dx = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.$$

Conclusion

$$\boxed{\int_0^1 \frac{\ln(1+x)}{x} dx = \frac{\pi^2}{12}}$$

Exercise 4

(a)-

$$\int_0^1 \sqrt{-\ln x} dx$$

Substitution

Let $u = -\ln x$. Then:

- When $x = 0$, $u \rightarrow +\infty$.
- When $x = 1$, $u = 0$.
- Differentiating: $du = -\frac{1}{x} dx$, so $dx = -x du = -e^{-u} du$.

The integral becomes

$$\int_0^1 \sqrt{-\ln x} dx = \int_{\infty}^0 \sqrt{u} \cdot (-e^{-u}) du = \int_0^{\infty} u^{1/2} e^{-u} du.$$

Recognize the Gamma Function

The integral $\int_0^{\infty} u^{1/2} e^{-u} du$ is the **Gamma function** evaluated at $\frac{3}{2}$:

$$\Gamma(z) = \int_0^{\infty} u^{z-1} e^{-u} du.$$

Here, $z = \frac{3}{2}$, so:

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2},$$

since $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

$$\boxed{\frac{\sqrt{\pi}}{2}}$$

(b)-

$$\int_0^{\pi/2} \sin^n x \, dx$$

for Positive Integers n

Reduction Formula

The integral $I_n = \int_0^{\pi/2} \sin^n x \, dx$ can be evaluated using the **Wallis reduction formula**:

$$I_n = \frac{n-1}{n} I_{n-2}.$$

The base cases are:

- $I_0 = \frac{\pi}{2},$
- $I_1 = 1.$

Separate Cases for Even and Odd n

- **For even $n = 2m$:**

$$I_{2m} = \frac{(2m-1)(2m-3)\cdots 1}{2m(2m-2)\cdots 2} \cdot \frac{\pi}{2} = \frac{(2m)!}{4^m(m!)^2} \cdot \frac{\pi}{2}.$$

Alternatively, using double factorials:

$$I_{2m} = \frac{(2m-1)!!}{(2m)!!} \cdot \frac{\pi}{2}.$$

- **For odd $n = 2m + 1$:**

$$I_{2m+1} = \frac{2m(2m-2)\cdots 2}{(2m+1)(2m-1)\cdots 1} = \frac{(2m)!!}{(2m+1)!!}.$$

1. $n = 2$:

$$I_2 = \frac{1}{2} I_0 = \frac{\pi}{4}.$$

2. $n = 3$:

$$I_3 = \frac{2}{3} I_1 = \frac{2}{3}.$$

For general n

$$\int_0^{\pi/2} \sin^n x \, dx = \begin{cases} \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2} & \text{if } n \text{ is even,} \\ \frac{(n-1)!!}{n!!} & \text{if } n \text{ is odd.} \end{cases}$$

YouTube

https://youtu.be/0aVVudlsq40?si=SIEKNGbH_rCalmDb

(c)-

$$\int_0^{\infty} x^2 e^{-x^2} \, dx$$

Method 1: Integration by Parts

Let's use integration by parts with:

- $u = x$ (so $du = dx$),
- $dv = x e^{-x^2} \, dx$ (so $v = -\frac{1}{2} e^{-x^2}$).

Then:

$$\int_0^{\infty} x^2 e^{-x^2} \, dx = -\frac{x}{2} e^{-x^2} \Big|_0^{\infty} + \frac{1}{2} \int_0^{\infty} e^{-x^2} \, dx.$$

The boundary term vanishes:

- At $x = 0$: $-\frac{0}{2} \cdot 1 = 0,$
- At $x \rightarrow \infty$: $-\frac{x}{2} e^{-x^2} \rightarrow 0$ (exponential decay dominates).

The remaining integral is the Gaussian integral:

$$\int_0^{\infty} e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}.$$

Thus:

$$\int_0^{\infty} x^2 e^{-x^2} \, dx = \frac{1}{2} \cdot \frac{\sqrt{\pi}}{2} = \boxed{\frac{\sqrt{\pi}}{4}}.$$

Method 2: Gamma Function

Recognize that:

$$\int_0^{\infty} x^n e^{-x^2} dx = \frac{1}{2} \Gamma\left(\frac{n+1}{2}\right).$$

For $n = 2$:

$$\int_0^{\infty} x^2 e^{-x^2} dx = \frac{1}{2} \Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \cdot \frac{\sqrt{\pi}}{2} = \boxed{\frac{\sqrt{\pi}}{4}}.$$

(d)-

$$\int_{-1}^1 \frac{x^3 + 2x^2 + 3x + 4}{(x^2 + 1)^2} dx$$

Simplify the Integrand

The numerator $(x^3 + 2x^2 + 3x + 4)$ is degree 3, while the denominator $(x^2 + 1)^2$ is degree 4, so we use partial fractions.

Partial Fraction Decomposition

Assume:

$$\frac{x^3 + 2x^2 + 3x + 4}{(x^2 + 1)^2} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{(x^2 + 1)^2}.$$

Multiply through by $(x^2 + 1)^2$

$$x^3 + 2x^2 + 3x + 4 = (Ax + B)(x^2 + 1) + Cx + D.$$

Expand and equate coefficients:

$$x^3 + 2x^2 + 3x + 4 = Ax^3 + Bx^2 + (A + C)x + (B + D).$$

Solving:

- $A = 1$,
- $B = 2$,
- $A + C = 3 \implies C = 2$,
- $B + D = 4 \implies D = 2$.

Thus:

$$\frac{x^3 + 2x^2 + 3x + 4}{(x^2 + 1)^2} = \frac{x + 2}{x^2 + 1} + \frac{2x + 2}{(x^2 + 1)^2}.$$

Integrate Term by Term

Split the integral:

$$\int_{-1}^1 \left(\frac{x}{x^2 + 1} + \frac{2}{x^2 + 1} + \frac{2x}{(x^2 + 1)^2} + \frac{2}{(x^2 + 1)^2} \right) dx.$$

1. **First Term:

$$\int_{-1}^1 \frac{x}{x^2 + 1} dx$$

- Odd function over symmetric limits:

$$\int_{-1}^1 \frac{x}{x^2 + 1} dx = 0.$$

2. Second Term:

$$\int_{-1}^1 \frac{2}{x^2 + 1} dx$$

- Standard integral:

$$2 \int_{-1}^1 \frac{1}{x^2 + 1} dx = 2 [\tan^{-1} x]_{-1}^1 = \pi.$$

3. **Third Term:**

$$\int_{-1}^1 \frac{2x}{(x^2+1)^2} dx$$

- Odd function:

$$\int_{-1}^1 \frac{2x}{(x^2+1)^2} dx = 0.$$

4. **Fourth Term:** $\int_{-1}^1 \frac{2}{(x^2+1)^2} dx$

- Using reduction formula:

$$\int \frac{1}{(x^2+1)^2} dx = \frac{x}{2(x^2+1)} + \frac{1}{2} \tan^{-1} x + C.$$

Evaluating:

$$2 \left[\frac{x}{2(x^2+1)} + \frac{1}{2} \tan^{-1} x \right]_{-1}^1 = 1 + \frac{\pi}{2}.$$

Add all non-zero terms:

$$0 + \pi + 0 + \left(1 + \frac{\pi}{2}\right) = \boxed{1 + \frac{3\pi}{2}}.$$

(e)-

$$\int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{\sin x + \cos x} dx$$

1. **Define the Integral**

Let

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{\sin x + \cos x} dx$$

2. **Apply the Symmetry Trick:**

Using the identity $\int_0^a f(x) dx = \int_0^a f(a-x) dx$, we rewrite I as:

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin\left(\frac{\pi}{2} - x\right) - \cos\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx$$

3. **Simplify Using Trigonometric Identities:**

- $\sin\left(\frac{\pi}{2} - x\right) = \cos x$
- $\cos\left(\frac{\pi}{2} - x\right) = \sin x$

Thus the integrand becomes:

$$\frac{\cos x - \sin x}{\cos x + \sin x} = - \left(\frac{\sin x - \cos x}{\sin x + \cos x} \right)$$

Establish the Relationship

Substituting back:

$$I = \int_0^{\frac{\pi}{2}} - \left(\frac{\sin x - \cos x}{\sin x + \cos x} \right) dx = -I$$

5. solve for I

$$I = -I \implies 2I = 0 \implies I = 0$$

Final Answer

$$\boxed{\int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{\sin x + \cos x} dx = 0}$$

Alternate Solution Using Substitution

1. **Let** $u = \sin x + \cos x$

Then $du = (\cos x - \sin x) dx = -(\sin x - \cos x) dx$

2. Change of Variables

- When $x = 0$, $u = 1$
- When $x = \frac{\pi}{2}$, $u = 1$

The integral becomes:

$$\int_1^1 \frac{-du}{u} = 0$$

(f)-

$$\int_0^{\pi} e^{-x} \sin x \, dx$$

3. Setup Integration by Parts

Let

- $u = \sin x \Rightarrow du = \cos x \, dx$
- $dv = e^{-x} dx \Rightarrow v = -e^{-x}$

Apply integration by parts:

$$\int e^{-x} \sin x \, dx = -e^{-x} \sin x + \int e^{-x} \cos x \, dx$$

4. Second Integration by Parts:

For $\int e^{-x} \cos x \, dx$, let:

- $u = \cos x \Rightarrow du = -\sin x \, dx$
- $dv = e^{-x} dx \Rightarrow v = -e^{-x}$

Thus

$$\int e^{-x} \cos x \, dx = -e^{-x} \cos x - \int e^{-x} \sin x \, dx$$

5. Combine Results

Let $I = \int e^{-x} \sin x \, dx$, then:

$$I = -e^{-x} \sin x - e^{-x} \cos x - I$$

Solving for I

$$2I = -e^{-x}(\sin x + \cos x)$$

$$I = -\frac{e^{-x}(\sin x + \cos x)}{2} + C$$

6. Evaluate Definite Integral

Compute from 0 to π :

$$\left[-\frac{e^{-x}(\sin x + \cos x)}{2} \right]_0^{\pi} = -\frac{e^{-\pi}(0 - 1)}{2} + \frac{e^0(0 + 1)}{2} = \frac{e^{-\pi}}{2} + \frac{1}{2}$$

7. Final Simplification

$$\frac{1 + e^{-\pi}}{2} = \frac{1}{2}(1 + e^{-\pi})$$

Final Answer:

$$\boxed{\frac{1 + e^{-\pi}}{2}}$$

(g)

$$I_n = \int_0^1 (\ln x)^n \, dx$$

Integration by parts setup:

$$u = (\ln x)^n \Rightarrow du = n(\ln x)^{n-1} \cdot \frac{1}{x} \, dx$$

$$dv = dx \Rightarrow v = x$$

Apply integration by parts:

$$I_n = x(\ln x)^n \Big|_0^1 - n \int_0^1 (\ln x)^{n-1} dx$$

Boundary term evaluation:

$$x(\ln x)^n \Big|_0^1 = 0$$

Recurrence relation obtained:

$$I_n = -nI_{n-1}$$

Base case:

$$I_0 = \int_0^1 1 dx = 1$$

First computations:

$$I_1 = -1$$

$$I_2 = 2$$

$$I_3 = -6$$

General solution pattern:

$$I_n = (-1)^n n!$$

Inductive verification:

$$\text{Base holds: } I_0 = (-1)^0 0! = 1$$

$$\text{Step holds: } I_{k+1} = (-1)^{k+1} (k+1)!$$

Final answer

$$\boxed{(-1)^n n!}$$

(h)

$$I = \int_0^\infty \frac{\ln x}{1+x^2} dx$$

Step 1 Use the substitution $x = \frac{1}{t}$

When $x \rightarrow 0^+$, $t \rightarrow \infty$

When $x \rightarrow \infty$, $t \rightarrow 0^+$

$$dx = -\frac{1}{t^2} dt$$

Step 2: Transform the integral

$$\begin{aligned} I &= \int_0^\infty \frac{\ln(1/t)}{1+(1/t)^2} \left(-\frac{1}{t^2}\right) dt \\ &= \int_0^\infty \frac{-\ln t}{1+\frac{1}{t^2}} \cdot \frac{1}{t^2} dt \\ &= -\int_0^\infty \frac{\ln t}{t^2+1} dt \end{aligned}$$

$$= -I$$

Solve for I

$$I = -I$$

$$2I = 0$$

$$I = 0$$

Verification

We can split the integral at $x=1$:

$$\int_0^1 \frac{\ln x}{1+x^2} dx + \int_1^\infty \frac{\ln x}{1+x^2} dx = 0$$

The substitution shows these two integrals are negatives of each other.

Final Answer

$$\boxed{0}$$

second method

Define the integral:

$$I = \int_0^\infty \frac{\ln x}{1+x^2} dx$$

Apply substitution $x = e^t$:

$$dx = e^t dt$$

$$x = 0 \Rightarrow t = -\infty$$

$$x = \infty \Rightarrow t = \infty$$

Transform the integral:

$$I = \int_{-\infty}^{\infty} \frac{t}{1 + e^{2t}} e^t dt$$

Simplify the integrand:

$$\frac{e^t}{1 + e^{2t}} = \frac{1}{e^{-t} + e^t} = \frac{1}{2 \cosh t}$$

Rewrite integral:

$$I = \frac{1}{2} \int_{-\infty}^{\infty} \frac{t}{\cosh t} dt$$

Observe symmetry:

The integrand $\frac{t}{\cosh t}$ is odd

Integral of odd function over symmetric limits:

$$\int_{-\infty}^{\infty} (\text{odd function}) dt = 0$$

Final conclusion:

$$I = 0$$

Final answer

$$\boxed{0}$$

(k)-

$$B(a+1, b+1) = \int_0^1 x^a (1-x)^b dx$$

Connection to Gamma functions:

$$B(a+1, b+1) = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)}$$

For integer arguments:

When $a, b \in \mathbb{Z}^+$

$$B(a+1, b+1) = \frac{a!b!}{(a+b+1)!}$$

Special case when $a = b$:

$$B(a+1, a+1) = \frac{\Gamma(a+1)^2}{\Gamma(2a+2)}$$

Alternative expression using binomial coefficients:

$$\int_0^1 x^a (1-x)^b dx = \frac{1}{(a+b+1) \binom{a+b}{a}}$$

Final answer in terms of Beta function

$$\boxed{B(a+1, b+1) = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)}}$$

Exercise 5

(a)-

$$I = \int_0^{\infty} \frac{x}{(x^2+1)^2} dx$$

Apply substitution:

Let

$$u = x^2 + 1$$

Then $du = 2x dx$

$$\Rightarrow \frac{du}{2} = x dx$$

Change limits:

When $x = 0$, $u = 1$

When $x = \infty$, $u = \infty$

Transform the integral:

$$I = \frac{1}{2} \int_1^{\infty} \frac{1}{u^2} du$$

Evaluate the integral:

$$I = \frac{1}{2} \left[-\frac{1}{u} \right]_1^{\infty}$$

Compute limits:

$$I = \frac{1}{2} (0 - (-1)) = \frac{1}{2}$$

Final answer

$$\boxed{\frac{1}{2}}$$

(b)- check kserra textbook normal distrubtion (Proof)

Exercise 6

$$F(\alpha) = \int_0^1 x^{\alpha-1} \ln x \, dx$$

Integration by parts setup:

Let

$$u = \ln x \Rightarrow du = \frac{1}{x} dx$$

$$dv = x^{\alpha-1} dx \Rightarrow v = \frac{x^{\alpha}}{\alpha}$$

Apply integration by parts:

$$F(\alpha) = \frac{x^{\alpha}}{\alpha} \ln x \Big|_0^1 - \int_0^1 \frac{x^{\alpha}}{\alpha} \cdot \frac{1}{x} dx$$

Evaluate boundary terms:

$$\lim_{x \rightarrow 0^+} x^{\alpha} \ln x = 0$$

for $\alpha > 0$

$$\frac{x^{\alpha}}{\alpha} \ln x \Big|_0^1 = 0 - 0 = 0$$

Simplify remaining integral:

$$F(\alpha) = -\frac{1}{\alpha} \int_0^1 x^{\alpha-1} dx$$

Evaluate the integral:

$$\int_0^1 x^{\alpha-1} dx = \frac{x^{\alpha}}{\alpha} \Big|_0^1 = \frac{1}{\alpha}$$

Final result:

$$F(\alpha) = -\frac{1}{\alpha^2}$$

Final answer:

$$\boxed{F(\alpha) = -\frac{1}{\alpha^2}} \text{ for } \alpha > 0$$

Exercise 7

$$I(a) = \int_0^{\infty} \frac{x^a}{e^x - 1} dx \quad \text{for } a > 0$$

Expand the Denominator as a Series

$$\frac{1}{e^x - 1} = e^{-x} \cdot \frac{1}{1 - e^{-x}} = e^{-x} \sum_{n=0}^{\infty} e^{-nx} = \sum_{n=1}^{\infty} e^{-nx}$$

Substitute into the Integral and Interchange Sum and Integral

$$I(a) = \int_0^{\infty} x^a \left(\sum_{n=1}^{\infty} e^{-nx} \right) dx = \sum_{n=1}^{\infty} \int_0^{\infty} x^a e^{-nx} dx$$

Evaluate the Inner Integral Using Substitution

Let $u = nx$, then $dx = \frac{du}{n}$

$$\int_0^{\infty} x^a e^{-nx} dx = \int_0^{\infty} \left(\frac{u}{n} \right)^a e^{-u} \cdot \frac{du}{n} = \frac{1}{n^{a+1}} \int_0^{\infty} u^a e^{-u} du$$

Recognize the Gamma Function

$$\int_0^{\infty} u^a e^{-u} du = \Gamma(a+1)$$

Thus

$$\int_0^{\infty} x^a e^{-nx} dx = \frac{\Gamma(a+1)}{n^{a+1}}$$

Sum the Series to Identify the Zeta Function

$$I(a) = \sum_{n=1}^{\infty} \frac{\Gamma(a+1)}{n^{a+1}} = \Gamma(a+1) \sum_{n=1}^{\infty} \frac{1}{n^{a+1}} = \Gamma(a+1) \zeta(a+1)$$

Final Answer

$$\boxed{\Gamma(a+1) \zeta(a+1)}$$

Exercise 8

(a)-

$$I = \int_0^{\infty} \frac{dx}{(1+e^x)(1+e^{-x})}$$

we will analyze its convergence or divergence.

Simplify the Integrand

First, simplify the denominator:

$$(1+e^x)(1+e^{-x}) = 1 \cdot 1 + 1 \cdot e^{-x} + e^x \cdot 1 + e^x \cdot e^{-x} = 1 + e^{-x} + e^x + 1 = 2 + e^x + e^{-x}.$$

notice that:

$$(1+e^x)(1+e^{-x}) = (1+e^x) \left(1 + \frac{1}{e^x} \right) = (1+e^x) \left(\frac{e^x + 1}{e^x} \right) = \frac{(1+e^x)^2}{e^x}.$$

Thus, the integrand can be rewritten as:

$$\frac{1}{(1+e^x)(1+e^{-x})} = \frac{e^x}{(1+e^x)^2}.$$

Substitute to Simplify Further

Let $u = 1 + e^x$, then $du = e^x dx$, and when $x = 0$, $u = 2$, and as $x \rightarrow \infty$, $u \rightarrow \infty$. The integral becomes

$$I = \int_2^{\infty} \frac{du}{u^2}.$$

Evaluate the Integral

The integral is now straightforward:

$$I = \int_2^{\infty} \frac{du}{u^2} = \left[-\frac{1}{u} \right]_2^{\infty} = 0 - \left(-\frac{1}{2} \right) = \frac{1}{2}.$$

Conclusion

The integral converges, and its value is:

$$\boxed{\frac{1}{2}}$$

(b)-Evaluate the improper integral:

$$I = \int_0^{\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx$$

Substitution to Simplify the Integral

Let's set:

$$u = \sqrt{x}$$

Then:

$$x = u^2 \Rightarrow dx = 2u du$$

Now, change the limits:

- When $x = 0$, $u = 0$.
- When $x \rightarrow \infty$, $u \rightarrow \infty$.

Substitute into the integral:

$$\int_0^{\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx = \int_0^{\infty} \frac{e^{-u}}{u} \cdot 2u du = 2 \int_0^{\infty} e^{-u} du$$

Step 2: Evaluate the Simplified Integral

The integral $\int_0^{\infty} e^{-u} du$ is a standard exponential integral:

$$\int_0^{\infty} e^{-u} du = [-e^{-u}]_0^{\infty} = 0 - (-1) = 1$$

Thus:

$$2 \int_0^{\infty} e^{-u} du = 2 \times 1 = 2$$

Final Answer then its **CV**

$$\boxed{2}$$

(c)- Evaluate the improper integral:

$$I = \int_0^1 \ln x dx$$

Identify the Improper Nature

The function $\ln x$ is undefined at $x = 0$ and approaches $-\infty$ as $x \rightarrow 0^+$.

Rewrite as a Limit

$$I = \lim_{a \rightarrow 0^+} \int_a^1 \ln x dx$$

Find the Antiderivative

$$\int \ln x dx = x \ln x - x + C$$

Evaluate the Definite Integral

$$\begin{aligned} \int_a^1 \ln x dx &= [x \ln x - x]_a^1 = (1 \cdot \ln 1 - 1) - (a \ln a - a) \\ &= (0 - 1) - (a \ln a - a) = -1 - a \ln a + a \end{aligned}$$

Take the Limit

$$\lim_{a \rightarrow 0^+} (-1 - a \ln a + a) = -1 - \lim_{a \rightarrow 0^+} a \ln a + \lim_{a \rightarrow 0^+} a$$

We know: $\lim_{a \rightarrow 0^+} a \ln a = 0$ and $\lim_{a \rightarrow 0^+} a = 0$

Thus: $I = -1 - 0 + 0 = -1$

The integral converges to: $\boxed{-1}$

(d)-

$$\int_1^{\infty} \frac{\ln x}{x^2} dx$$

Identify the Improper Nature

The integral is improper because

1. The upper limit is infinity
2. The integrand $\frac{\ln x}{x^2}$ is well-defined and continuous on $[1, \infty)$

Rewrite as a Limit

Express the integral as

$$\lim_{b \rightarrow \infty} \int_1^b \frac{\ln x}{x^2} dx$$

Integration by Parts

- Let $u = \ln x \Rightarrow du = \frac{1}{x} dx$
- Let $dv = \frac{1}{x^2} dx \Rightarrow v = -\frac{1}{x}$

Then

$$\int \frac{\ln x}{x^2} dx = -\frac{\ln x}{x} + \int \frac{1}{x^2} dx = -\frac{\ln x}{x} - \frac{1}{x} + C$$

Evaluate the Definite Integral

Now evaluate from 1 to b:

$$\begin{aligned} \int_1^b \frac{\ln x}{x^2} dx &= \left[-\frac{\ln x}{x} - \frac{1}{x} \right]_1^b = \left(-\frac{\ln b}{b} - \frac{1}{b} \right) - \left(-\frac{\ln 1}{1} - \frac{1}{1} \right) \\ &= -\frac{\ln b}{b} - \frac{1}{b} + 1 \end{aligned}$$

Take the Limit as $b \rightarrow \infty$

$$\lim_{b \rightarrow \infty} \left(-\frac{\ln b}{b} - \frac{1}{b} + 1 \right) = 0 - 0 + 1 = 1$$

We used:

1. $\lim_{b \rightarrow \infty} \frac{\ln b}{b} = 0$ (by L'Hôpital's rule)
 2. $\lim_{b \rightarrow \infty} \frac{1}{b} = 0$
- Step 6: Convergence Analysis
- The integral converges because:
3. For large x , $\frac{\ln x}{x^2} \ll \frac{1}{x^{1.5}}$ (since any power > 1 dominates $\ln x$)
 4. $\int_1^{\infty} \frac{1}{x^p} dx$ converges for $p > 1$

Final Answer

The integral converges to:

$$\boxed{1}$$

Exercise 9

Evaluating the Integral

$$I = \int_0^{\infty} \frac{\sin x}{x} dx$$

Part (a): Show that the integral $I = \int_0^{\infty} \frac{\sin x}{x} dx$ is convergent.

Approach:

We split the integral into two parts to analyze convergence:

1. **Integral from 0 to 1:**

- Near $x = 0$, $\frac{\sin x}{x}$ has a removable singularity since $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.
- Thus, $\frac{\sin x}{x}$ is continuous and bounded on $[0, 1]$, making $\int_0^1 \frac{\sin x}{x} dx$ finite.

2. **Integral from 1 to ∞ :**

- Use integration by parts:

$$\text{Let } u = \frac{1}{x}, dv = \sin x dx \Rightarrow du = -\frac{1}{x^2} dx, v = -\cos x.$$

Then:

$$\int_1^\infty \frac{\sin x}{x} dx = -\frac{\cos x}{x} \Big|_1^\infty - \int_1^\infty \frac{\cos x}{x^2} dx$$

- The boundary term evaluates to:

$$\lim_{x \rightarrow \infty} -\frac{\cos x}{x} + \cos 1 = 0 + \cos 1 = \cos 1.$$

- The remaining integral $\int_1^\infty \frac{\cos x}{x^2} dx$ converges absolutely since $|\frac{\cos x}{x^2}| \leq \frac{1}{x^2}$ and $\int_1^\infty \frac{1}{x^2} dx$ converges.

Both parts are finite, so I converges.

Define sequences I_n and J_n .

Given

$$I_n = \int_0^{\pi/2} \frac{\sin((2n+1)x)}{\sin x} dx, \quad J_n = \int_0^{\pi/2} \frac{\sin((2n+1)x)}{x} dx$$

i) Show that $J_n \rightarrow I$ using substitution $t = (2n+1)x$.

$$\text{Let } t = (2n+1)x \Rightarrow dt = (2n+1)dx.$$

$$\text{When } x = 0, t = 0; \text{ when } x = \frac{\pi}{2}, t = (2n+1)\frac{\pi}{2}.$$

Substitute into J_n :

$$J_n = \int_0^{(2n+1)\pi/2} \frac{\sin t}{\frac{t}{2n+1}} \cdot \frac{dt}{2n+1} = \int_0^{(2n+1)\pi/2} \frac{\sin t}{t} dt$$

As $n \rightarrow \infty$, $(2n+1)\frac{\pi}{2} \rightarrow \infty$, so:

$$\lim_{n \rightarrow \infty} J_n = \int_0^\infty \frac{\sin t}{t} dt = I.$$

ii) Show that $I_n - I_{n-1} = 0$ and deduce $I_n = \frac{\pi}{2}$.

Compute $I_n - I_{n-1}$:

$$I_n - I_{n-1} = \int_0^{\pi/2} \left(\frac{\sin((2n+1)x)}{\sin x} - \frac{\sin((2n-1)x)}{\sin x} \right) dx.$$

Using the identity $\sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$:

$$\sin((2n+1)x) - \sin((2n-1)x) = 2 \cos(2nx) \sin x.$$

Thus:

$$I_n - I_{n-1} = \int_0^{\pi/2} \frac{2 \cos(2nx) \sin x}{\sin x} dx = 2 \int_0^{\pi/2} \cos(2nx) dx.$$

The integral of $\cos(2nx)$ over $[0, \pi/2]$ is:

$$\int_0^{\pi/2} \cos(2nx) dx = \frac{\sin(2nx)}{2n} \Big|_0^{\pi/2} = \frac{\sin(n\pi)}{2n} - 0 = 0.$$

Therefore:

$$I_n - I_{n-1} = 0 \quad \forall n \in \mathbb{N}^*.$$

This implies I_n is constant. Compute I_0 :

$$I_0 = \int_0^{\pi/2} \frac{\sin x}{\sin x} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}.$$

Thus:

$$I_n = \frac{\pi}{2} \quad \forall n \in \mathbb{N}.$$

Part (c) Analyze $f(x) = \frac{1}{\sin x} - \frac{1}{x}$. **

i) Show $f(x)$ extends continuously to 0 as a C^1 function on $[0, \pi/2]$

First, compute $\lim_{x \rightarrow 0^+} f(x)$:

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{\sin x} - \frac{1}{x} \right) = \lim_{x \rightarrow 0^+} \frac{x - \sin x}{x \sin x}.$$

Using Taylor expansions ($\sin x \approx x - \frac{x^3}{6}$ for small x):

$$\frac{x - \sin x}{x \sin x} \approx \frac{x - (x - \frac{x^3}{6})}{x^2} = \frac{\frac{x^3}{6}}{x^2} = \frac{x}{6} \rightarrow 0.$$

Thus, defining $f(0) = 0$ makes f continuous at 0.

Next, compute $f'(x)$:

$$f'(x) = -\frac{\cos x}{\sin^2 x} + \frac{1}{x^2}.$$

Compute $\lim_{x \rightarrow 0^+} f'(x)$ using Taylor expansions ($\cos x \approx 1 - \frac{x^2}{2}$, $\sin x \approx x - \frac{x^3}{6}$):

$$f'(x) \approx -\frac{1 - \frac{x^2}{2}}{x^2 \left(1 - \frac{x^2}{6}\right)^2} + \frac{1}{x^2} \approx -\frac{1}{x^2} \left(1 - \frac{x^2}{2}\right) \left(1 + \frac{x^2}{3}\right) + \frac{1}{x^2}.$$

Expanding and simplifying:

$$f'(x) \approx -\frac{1}{x^2} \left(1 + \frac{x^2}{3} - \frac{x^2}{2} - \frac{x^4}{6}\right) + \frac{1}{x^2} = -\frac{1}{x^2} + \frac{1}{6} + \frac{1}{x^2} = \frac{1}{6}.$$

Thus, $f'(0) = \frac{1}{6}$, and f is C^1 on $[0, \pi/2]$.

ii) Deduce $\lim_{n \rightarrow \infty} (I_n - J_n) = 0$ and $I = \frac{\pi}{2}$.

We have:

$$I_n - J_n = \int_0^{\pi/2} \sin((2n+1)x) \left(\frac{1}{\sin x} - \frac{1}{x} \right) dx = \int_0^{\pi/2} \sin((2n+1)x) f(x) dx.$$

Since f is C^1 on $[0, \pi/2]$, the Riemann-Lebesgue lemma implies:

$$\lim_{n \rightarrow \infty} \int_0^{\pi/2} \sin((2n+1)x) f(x) dx = 0.$$

But $\lim_{n \rightarrow \infty} J_n = I$ and $I_n = \frac{\pi}{2}$ for all n , so:

$$\frac{\pi}{2} - I = 0 \implies I = \frac{\pi}{2}.$$

$$I = \boxed{\frac{\pi}{2}}$$

Exercise 10

(a)

$$\int_{\pi}^{\infty} \frac{\cos x}{\sqrt{x}} dx$$

is **conditionally convergent**, meaning

1. The integral **converges** (i.e., the limit $\int_{\pi}^{\infty} \frac{\cos x}{\sqrt{x}} dx$ exists).
2. The integral **does not converge absolutely** (i.e., $\int_{\pi}^{\infty} \frac{\cos x}{\sqrt{x}} dx$).

Convergence of the Integral (Conditional Convergence)

Substitution

Let $u = \sqrt{x}$. Then

$$x = u^2, \quad dx = 2u du.$$

The integral becomes:

$$\int_{\sqrt{\pi}}^{\infty} \frac{\cos u}{u^2} \cdot 2u \, du = 2 \int_{\sqrt{\pi}}^{\infty} \frac{\cos u}{u} \, du.$$

Analysis of $\int_a^{\infty} \frac{\cos u}{u} \, du$

This is a classic **Dirichlet-type integral**, which converges conditionally. To verify:

3. Oscillatory Nature The numerator $\cos u$ oscillates between -1 and 1 .
4. Decay of the Denominator The denominator u ensures the amplitude of oscillations decays like $\frac{1}{u}$.
5. Dirichlet Test for Integrals:

Let $f(u) = \cos u$ (bounded, with $|\int_a^b \cos u \, du| = |-\sin(u)|_a^b| = |\sin(b) - \sin(a)| \leq |\sin(b)| + |\sin(a)| \leq 1 + 1 = 2$ for any a, b).

Let $g(u) = \frac{1}{u}$ (monotonically decreasing to 0 as $u \rightarrow \infty$ for $u > 0$).

By the Dirichlet test, $\int_a^{\infty} \frac{\cos u}{u} \, du$ converges for any $a > 0$. Since $\sqrt{\pi} > 0$, $\int_{\sqrt{\pi}}^{\infty} \frac{\cos u}{u} \, du$ converges.

Thus, the original integral converges.

Divergence of the Absolute Integral (Non-Absolute Convergence)

Consider the absolute version

$$\int_{\pi}^{\infty} \left| \frac{\cos \sqrt{x}}{x} \right| dx = \int_{\pi}^{\infty} \frac{|\cos \sqrt{x}|}{x} dx.$$

Let $u = \sqrt{x}$, so $dx = 2u \, du$, and the integral becomes:

$$2 \int_{\sqrt{\pi}}^{\infty} \frac{|\cos u|}{u} \, du.$$

Lower Bound for Divergence

Since $|\cos u| \geq \cos^2 u = \frac{1+\cos 2u}{2}$, we have:

$$\frac{|\cos u|}{u} \geq \frac{1 + \cos 2u}{2u}.$$

Thus

$$2 \int_{\sqrt{\pi}}^{\infty} \frac{|\cos u|}{u} \, du \geq \int_{\sqrt{\pi}}^{\infty} \frac{1 + \cos 2u}{u} \, du.$$

Now

6. The integral $\int_{\sqrt{\pi}}^{\infty} \frac{1}{u} \, du$ diverges (logarithmic divergence, since $\int \frac{1}{u} \, du = \ln|u|$ and $\lim_{b \rightarrow \infty} \ln(b) = \infty$).
7. The integral $\int_{\sqrt{\pi}}^{\infty} \frac{\cos 2u}{u} \, du$ converges (by the Dirichlet test, similar to Step 1 with $f(u) = \cos(2u)$ and $g(u) = 1/u$).

Since one part diverges and the other converges, the overall integral on the right-hand side diverges. By the comparison test, the integral of the absolute value also diverges:

$$\int_{\sqrt{\pi}}^{\infty} \frac{|\cos u|}{u} \, du = \infty.$$

The integral $\int_{\pi}^{\infty} \frac{\cos x}{\sqrt{x}} \, dx$ is conditionally convergent.

(b)-

$$\int_{-1}^{\infty} \cos(x^2) \, dx$$

8. Integral Decomposition The integral is split due to the substitution $u = x^2$:

$$\int_{-1}^{\infty} \cos(x^2) \, dx = \int_{-1}^0 \cos(x^2) \, dx + \int_0^{\infty} \cos(x^2) \, dx$$

The first part $\int_{-1}^0 \cos(x^2) \, dx$ is an integral of a continuous function over a finite interval, so it converges absolutely. The convergence of the original integral depends on the convergence of $\int_0^{\infty} \cos(x^2) \, dx$.

9. **Substitution** $u = x^2$ for the main part:

$$\int_0^{\infty} \cos(x^2) \, dx = \frac{1}{2} \int_0^{\infty} \frac{\cos(u)}{\sqrt{u}} \, du$$

10. Analyzing $\int_0^{\infty} \frac{\cos(u)}{\sqrt{u}} \, du$: This improper integral has issues at both endpoints (0 and ∞). It's split into two parts

$$\int_0^{\infty} \frac{\cos(u)}{\sqrt{u}} \, du = \int_0^1 \frac{\cos(u)}{\sqrt{u}} \, du + \int_1^{\infty} \frac{\cos(u)}{\sqrt{u}} \, du$$

- **Behavior near 0:** $\int_0^1 \frac{\cos(u)}{\sqrt{u}} du$ converges absolutely because $\frac{\cos(u)}{\sqrt{u}} \sim \frac{1}{\sqrt{u}}$ as $u \rightarrow 0$, and $\int_0^1 \frac{1}{\sqrt{u}} du$ converges.
- **Behavior near ∞ :** $\int_1^\infty \frac{\cos(u)}{\sqrt{u}} du$ is analyzed using Dirichlet's Test:
 - $\int \cos(u) du = \sin(u)$ is bounded.
 - $g(u) = \frac{1}{\sqrt{u}}$ decreases monotonically to 0 as $u \rightarrow \infty$.
 Thus, $\int_1^\infty \frac{\cos(u)}{\sqrt{u}} du$ converges by Dirichlet's Test.

11. Absolute Convergence Check for $\int_1^\infty \frac{\cos(u)}{\sqrt{u}} du$

$$\int_1^\infty \left| \frac{\cos(u)}{\sqrt{u}} \right| du \geq \int_1^\infty \frac{\cos^2(u)}{\sqrt{u}} du = \frac{1}{2} \int_1^\infty \frac{1}{\sqrt{u}} du + \frac{1}{2} \int_1^\infty \frac{\cos(2u)}{\sqrt{u}} du$$

Since $\int_1^\infty \frac{1}{\sqrt{u}} du$ diverges, the integral of the absolute value diverges.

Conclusion for Integral $\int_{-1}^\infty \cos(x^2) dx$ is conditionally convergent

(c)-

$$\int_\pi^\infty x^2 \sin(x^4) dx$$

1. **Substitution $u = x^4$

$$\int_\pi^\infty x^2 \sin(x^4) dx = \frac{1}{4} \int_{\pi^4}^\infty \frac{\sin(u)}{u^{1/4}} du$$

2. **Analyzing $\int_{\pi^4}^\infty \frac{\sin(u)}{u^{1/4}} du$:** This improper integral is analyzed using Dirichlet's Test:

- $\int \sin(u) du = -\cos(u)$ is bounded.
 - $g(u) = \frac{1}{u^{1/4}}$ decreases monotonically to 0 as $u \rightarrow \infty$ for $u \geq \pi^4$.
- Thus, $\int_{\pi^4}^\infty \frac{\sin(u)}{u^{1/4}} du$ converges by Dirichlet's Test.

3. **Absolute Convergence Check for $\int_{\pi^4}^\infty \frac{\sin(u)}{u^{1/4}} du$:**

$$\int_{\pi^4}^\infty \left| \frac{\sin(u)}{u^{1/4}} \right| du \geq \frac{1}{2} \int_{\pi^4}^\infty \frac{1 - \cos(2u)}{u^{1/4}} du = \frac{1}{2} \int_{\pi^4}^\infty \frac{1}{u^{1/4}} du - \frac{1}{2} \int_{\pi^4}^\infty \frac{\cos(2u)}{u^{1/4}} du$$

Since $\int_{\pi^4}^\infty \frac{1}{u^{1/4}} du$ diverges (as $1/4 \leq 1$), the integral of the absolute value diverges.

Conclusion for Integral $\int_\pi^\infty x^2 \sin(x^4)$ is conditionally convergent